# Multi-modal Topology-embedded Graph Learning for Spatially Resolved Genes Prediction from Pathology Images with Prior Gene Similarity Information

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## Abstract

*The rapid development of spatial transcriptomics (ST) allows researchers to measure the spatial-level gene expression in tissues. Although powerful, the cost for collecting the ST data is expensive, and thus several studies aim to predict gene expression in ST by utilizing their corresponding H/E stained pathology images. The existing ST based gene expression prediction models either adopt the pre-trained networks or rely on the handcrafted features to describe the pathology images, which still lack a systematic way to combine them together to define a spot-level representation that can reflect the topological profiles of different spots. On the other hand, all the ST based gene prediction models treat the prediction task for each gene independently, which overlook the fact that the exploration of potential interrelationships among them can help improve the prediction performance for individual genes. To address the above issues, we propose a multi-modal topology-embedded graph learning algorithm guided by prior Gene Ontology similarity information (i.e., M2TGLGO) to predict the spatial resolved genes from pathology images. Specifically, M2TGLGO co-learns the image representation of different spots from both deep and handcrafted features by considering the within-modal and inter-modal interactions. Next, to keep the topological structure among different spots, a spatial-oriented ranking module is also incorporated to preserve their neighborhood similarity information. Finally, we present a Gene Ontology knowledge guided graph neural network for simultaneously predicting multiple gene expressions by considering their functional associations. We evaluate our method on three public available ST datasets, the experimental results show the effectiveness of our M2TGLGO in comparison with the existing studies.*

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Code and Dataset is available at: github.com/shihangjs/M2TGLGO

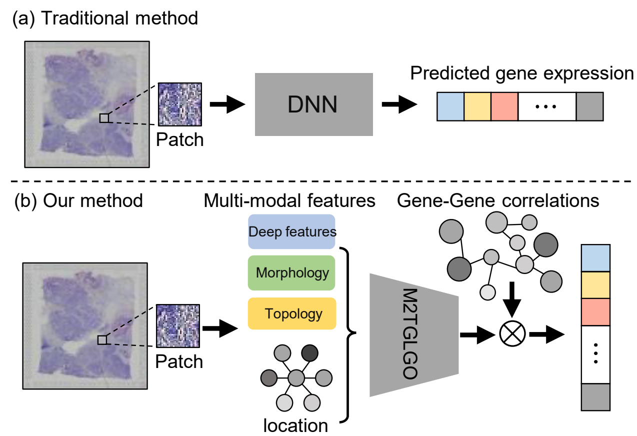


Figure 1. (a) Traditional ST based genes prediction models rely on single type of image feature (*e.g.,* deep feature) and treat the prediction task on each gene independently. (b) Our M2TGLGO combines multi-modal image features for the spatial resolved genes prediction by considering their biological similarities.

## 1. Introduction

Spatial transcriptomics (ST) is a cutting-edge technology that enables the measurement of gene expression at spatial resolution, making it possible to examine how gene expression varies across different spots as well as offering insights into the relationships between molecular activity and tissue structure [16][30]. Although powerful, the high cost of collecting ST data limits its widespread use in clinical applications [35][44][28], while their matched pathology images are cheap and easy to acquire. Hence, it has become a research hotspot to link the tissue structure of pathology images with the gene expression patterns that can help researchers infer molecular information of ST data in a more cost-effective and scalable way [39][43].

Recently, the rapid advancements in artificial intelligence and image-based technologies have shown their great potential in healthcare research [25], and several algorithms have been proposed to predict gene expression of ST from pathology images. A standard ST-based workflow for gene prediction starts with finding a suitable way to represent different spots from pathology images, followed by designing gene expression predictors to assess their corresponding molecular signatures. Generally, the existing image-based spot descriptors can be categorized into handcrafted and deep features, where the handcrafted features explore the morphology and density information of nuclei in each spot [11] [14] that can reflect the complex tumor micro-environment (TME) information in tissues [31]. On the other hand, the deep features extracted from the intermediate layers of a pre-trained network (*e.g.,* Resnet101 and Resnet18) [39] [5] is capable of capturing both low-level details (*e.g.,* edges, textures) and high-level concepts (*e.g.,* shapes, objects) from the pathology images.

Based on the learned representations from the pathology images, several methods have been presented to predict the gene expression in ST data. For instance, ST-Net [14] is the first study reporting the development of a deep learning algorithm for the prediction of local gene expression from H/E stained pathological images. The THItoGene model [18] is developed to systematically predict RNA-Seq profiles from whole-slide images by utilizing dynamic convolutional and capsule networks. In recent years, we also witness a bunch of studies that integrate both pathology images and spatial information for predicting the gene expression in ST data [27] [42] [40]. These studies model the intricate relationships among spatial locations via the graph neural network (GNN) or vision transformer (ViT) that are also effective for accomplishing the gene prediction task.

Although much progress has been achieved, most of the existing studies either rely on the handcrafted or deep features extracted from pathology images for gene prediction, it is still challenging to integrate them together with the spatial information for the representation of different spots. In addition, the existing GNN-based gene prediction models overlook the topological structure among different spots that may suffer from the over-smoothing problem. In such cases, the nodes tend to share similar representations [21], leading to the descriptors of different spots becoming indistinguishable. Last but not least, to the best of our knowledge, all the ST-based gene prediction models are constructed on the independent parallel hypothesis where the prediction task for each gene is treated independently and the important functional similarity among different genes are missed. Actually, research in biology suggests that the genes with analogous functions usually show similar expression patterns. It can be expected that better prediction performance can be achieved if the developed model takes the prior gene-gene association information into account.

Based on the above consideration, we propose a multi-modal topology-embedded graph learning algorithm guided by prior Gene Ontology knowledge (*i.e.,* M2TGLGO) to predict the spatial resolved gene expression from pathology images (shown in Fig. 1). We conduct experiments on three ST datasets and the experimental results demonstrate the superiority of our method when comparing with the existing studies. We summarize the main contributions of this study as follows:

1) We propose a multi-modal graph embedding (MMGE) algorithm that can co-learn the spot representations from both handcrafted and deep features by considering their within-modal and inter-modal interactions.

2) We develop a novel spatial-oriented neighborhood ranking module (SONRM) to preserve the topological structure of spots, where the neighborhood nodes that are spatially closer to the target spot should have more similar representations than other nodes.

3) We for the first time present a prior Gene Ontology (GO) knowledge guided GNN model for the prediction of different genes on ST data by considering their function similarity.

## 2. Related Work

**Descriptors for Pathology Image:** Generally, the existing pathology image descriptors can be divided into two categories, *i.e.,* handcrafted features and deep features. On one hand, the handcrafted descriptors focus on extracting nuclei morphological and distribution information from the pathology image that are with desired interpretability [3, 4, 11]. On the other hand, with the rapid development of the deep learning technology, extracting features from a pre-trained network (*e.g.,* ResNet101) has drawn much attentions [33, 41] since it offers a powerful starting point for the downstream pathology image analysis task. Based on the pre-trained features, Shao *et al* [32] propose a tumor micro-environment interactions guided graph embedding algorithm to learn the representations of image patches, Chan [2] present a heterogeneous graph representation learning algorithm that can capture the local structure of the whole-slide pathological images. Other studies include [20] developing an implicit neural representation model that can help detect the complex patterns found in pathology images. To the best of our knowledge, only the study in [19] combines the handcrafted and deep features to represent pathology images. However, it simply integrates the multi-modal features without considering the multi-modal interactions. Also, it works on the whole-slide pathological images that overlooks the spatial information among different spots for representation learning.

**Spatial Resolved Gene Prediction from Pathology Image:** In this field, ST-Net [14] and HE2RNA [29] are two pioneering studies that measure the relationship between pathology images and the spatial resolved gene expression data. Xie *et al* [39] present a bi-modal em-

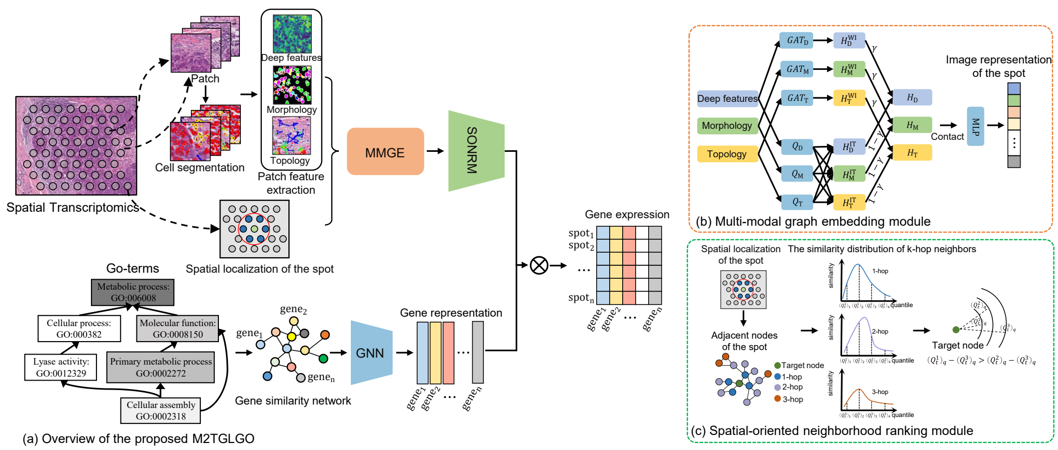


Figure 2. The flowchart of the proposed M2TGLGO framework. M2TGLGO starts by extracting both deep and handcrafted (*i.e.,* morphology and topology) image features in each spot of the ST data. Then, a multi-modal graph embedding algorithm(*i.e.,* MMGE) is proposed to fuse them effectively. Next, we present a spatial-oriented neighborhood ranking module (*i.e.,* SONRM) to keep the topological structure among different spots. Finally, a Gene Ontology(GO) knowledge guided graph neural network is proposed to predict different gene expressions by considering their functional associations.

bedding framework for predicting spatial transcriptomics from pathology images. THItoGene [18] is a hybrid neural network that utilizes dynamic convolutional and capsule networks to adaptively sense potential molecular signals in pathology images. In addition, Chung [5] propose a simple but effective spatial transcriptomics prediction model based on the multi-resolution features extracted from the pathology images. Moreover, for the purpose of incorporating the spatial positional information of different spots during model training, Hist2ST [42], HisToGene [27] and EGGN [40] are developed where the spatial relations of different spots are captured through Transformer or graph neural network modules. However, these graph learning based ST analysis methods rely on single type of pathology image feature and also overlook the topological structure among different spots that may lead to over-smoothing problem.

## 3. Method

We show the overall flowchart of our proposed M2TGLGO in Fig. 2, which is consisted of four steps. We firstly extract multi-modal histopathological image features from different spots in ST data. Then, a multi-modal graph embedding (MMGE) algorithm is presented to combine these different types of image feature together. To preserve the topological structure of different spots, a spatial-oriented neighborhood ranking module is also involved in our M2TGLGO. Finally, we leverage the prior Gene Ontology (GO) knowledge to calculate the function similarity among different genes, which is then applied to construct a graph neural network for gene prediction on ST data.

### 3.1. Pre-processing of the ST data.

In comparison with the Whole-slide pathological images (WSIs), the Spatial Transcriptomics (ST) data is characterized by its compilation of numerous spot images within a single slide, each accompanied by corresponding gene expression values. For each slide in different ST datasets used in this study, we firstly exclude spots located outside the primary tissue region. Then, we follow the study in [7] that applies the SCANPY package to perform logarithmic transformation and normalization on the gene expression data in order to identify a common set of 3000 genes with the highest variation for all the spots in each slide. For the matched pathological image, we crop it into patches according to the spatial context of the spots.

### 3.2. Image Feature Extraction for Spots.

For each spot image, we extract both deep and handcrafted features from it aimed at incorporating richer information for the downstream gene expression prediction task.
**Deep Feature:** For each spot image, we firstly uniformly adjust its size to $224 \times 224$, followed by applying the Resnet-101 network for deep feature extraction.
**Handcrafted Feature:** As to the handcrafted features, we pay more attention to the morphological and topological information of nucleus revealed in the pathological images. Specifically, for each spot image, we firstly employ HoVer-Net[12], which is pre-trained on the PanNuke dataset[9], to segment all the nuclei in it. Then, we extract *14 morphology features* for each nucleus including its area, length and shape information (Details are listed in the *Supplementary Materials*). In addition, we follow the study in [4] that adopts the Delaunay triangulation graph

to calculate *3 topological features* regarding the mean, maximum, and minimum distances from each nucleus to its neighboring nuclei. For both morphology and topology features, we aggregate the numerous nuclei-level features extracted from a spot into spot-level representation by calculating their average values.

### 3.3. Multi-modal Graph Embedding for the Representation of Spots

Since gene expression often displays local patterns, which are typically manifested in the histology images [26], it is necessary to explicitly model the spatial location information when predicting gene expression. Based on the above consideration, we develop a multi-modal graph embedding algorithm (MMGE) that can co-learn the spot representations from both handcrafted and deep features by considering their within-modal and inter-modal interactions (shown in Fig. 2(b)). Before we give details about the proposed framework, we firstly introduce some notations and we will use them in the remaining of this study.

Mathematically, given the slide with $N$ spots, we first construct the graph $G$ by adopting the K-nearest neighbor algorithm. Here, we set the number of nearest neighbors as 6. The inputs of our MMGE module are the adjacent matrix $A \in R^{N\times N}$ and the initialized multi-modal features matrix $X_m = [x_m^1, x_m^2, ..., x_m^N] \in R^{N\times d_m}$ where $m \in \{D, M, T\}$ represents the deep features, handcrafted morphology and topology features with the dimensionality of $d_m$, respectively. For the purpose of capturing the within-modal interactions among different spots, the graph attention networks (GATs)[36] is employed and can be formulated as:

$$h_m^i = \sigma(\sum_{j\in N_{(i)}} \alpha_m^{ij} W_m x_m^j) \tag{1}$$

Where $W_m \in \mathbb{R}^{d_m\times d^o}$ is a learnable weight matrix correspond to modality $m$, $\sigma$ is the non-linear Sigmoid function, $N_{(i)}$ denotes the set of neighbors of spot $i$, and $\alpha_m^{ij}$ is the attention coefficient representing the association between spot $i$ and $j$. As shown in Eq.(1), the updated representation of spot $i$ (*i.e.,* $h_m^i$) in modality $m$ can be treated as the weighted sum of its neighborhood spots, allowing a more flexible way for modeling the within-modal interactions among different nodes. Then, the within-modal embeddings for modality $m$ can be represented as $H_m^{\text{WI}} = [h_m^1, h_m^2, \ldots, h_m^N]^T \in R^{N\times d^o}$.

On the other hand, to model the inter-modal interactions among multi-modal data, we perform a similar aggregation way as we do for the calculation of within-modal interactions. Specifically, to ensure the consistency of dimensions among multi-modal data for future fusion, we firstly introduce a project matrix $Q_m \in R^{d_m\times d^o}$ for each modality denoted below:

$$X'_m = Relu(X_m Q_m) \tag{2}$$

We then calculate a similarity score between modalities $m$ and $n$ to assess their interactions as follows:

$$\beta_{m,n} = \frac{\exp(tr((X'_m)(X'_n)^T))}{\sum_{k\in\{D,M,T\}} \exp(tr((X'_m)(X'_k)^T))} \tag{3}$$

Where *tr(,)* indicates the trace operator. Then, the inter-modal aggregation for modality $m$ can be formulated as:

$$H_m^{\text{IT}} = \sum_{k\in\{D,M,T\}} \beta_{m,k} X'_k \tag{4}$$

Next, for the purpose of acquiring the modality-specific spots representations with both neighboring nodes information and the complementary knowledge derived from other modalities, we combine the within-modal embeddings with the inter-modal embeddings as follows:

$$H_m = \gamma H_m^{\text{WI}} + (1-\gamma) H_m^{\text{IT}} \tag{5}$$

where $\gamma \in [0, 1]$ is a regularization parameter controlling the relative contribution between within-modal and inter-modal embeddings. Finally, the representations of spots $H = [h_1, h_2, \ldots, h_N] \in R^{N\times v_s}$ can be obtained by combining all the modality-specific embeddings:

$$H = \sigma\left(W_S \cdot \text{Concat}_{m\in\{D,M,T\}} H_m\right) \tag{6}$$

Where $W_S$ is a learnable parameter and $\sigma(.)$ is an element-wise Sigmoid activation function.

### 3.4. Preserving the Topological Structure of Spots via Spatial-Oriented Neighborhood Ranking.

Although the multi-modal graph embedding algorithm (MMGE) introduced before can learn spots representations by considering their spatial association, it may bring the over-smoothing problem [21]. This is because the repeated aggregation of neighboring node features in each GAT layer causes the node representations to converge toward similar values, which reduces the model's ability to differentiate among spots. To address it, we firstly add a reconstruction loss with the aim of recovering the adjacent matrix $A$ from the learned representations $H$. Specifically, for nodes $i$ and $j$, the pairwise representation similarity is denoted as:

$$A'_{ij} = \sigma(\frac{h_i^\top h_j}{\|h_i\|\|h_j\|}) \tag{7}$$

Where $\sigma(.)$ is the Sigmoid function that enforce the value of $A'_{ij}$ ranging from 0 to 1. Then, the reconstruction loss can be formulated as:

$$\mathcal{L}_{\text{recon}} = \frac{1}{N^2}\sum_{i=1}^{N}\sum_{j=1}^{N}(A_{ij} - A'_{ij})^2 \tag{8}$$

Where $A$ is the adjacent matrix. As shown in Eq.(8), we hope that the directly connected nodes (*i.e.,* $A_{ij} = 1$) share similar features while the unconnected nodes (*i.e.,* $A_{ij} = 0$) come with different representations that can help relieve the over-smoothing problem to some extent. However, it only applies the direct neighbors without considering that the long-range correlations between nodes can also reflect the structure of the graph. Intuitively, for the purpose of preserving the topological profiles of the spots, the pairwise representation similarity between distant nodes should be lower than that among direct nodes. Consequently, we further develop a spatial-oriented neighborhood ranking module (SONRM) to satisfy such similarity constrains. More specifically, given any target node $t$ from the node set of graph $G$, we firstly extract its neighbors with different hop ranges (*i.e.,* 1-hop, 2-hop and 3-hop). For all the neighborhood nodes in specific range, we individually calculate their representation similarity $RS_t^r(r = 1, 2, 3)$ with the target node according to Eq.(7), and then compute the quartiles for each $RS_t^r$ denoted as $(Q_t^r)_q(q = 1, 2, 3, 4)$. Then, to keep the neighborhood similarity orders, we add the regularization term below to ensure that $(Q_t^r)_q$ with larger hop ranges should be with smaller value for each quartile.

$$\begin{aligned} L_s = \sum_{t\in G}\sum_{q=1}^{4} &\max(0, (Q_t^2)_q - (Q_t^1)_q + \epsilon) \\ &+ \max(0, (Q_t^3)_q - (Q_t^2)_q + \epsilon) \end{aligned} \tag{9}$$

Where $\epsilon$ is a small positive value to regulate the difference and we fix it as $0.01$ in this study.

### 3.5. Gene Ontology Knowledge Guided Graph Neural Network for Gene Predictions

After deriving the representations of different spots, the existing ST based studies [5, 17, 39] treat the prediction tasks on each gene independently, which overlook exploring the important biological similarity information among different genes. As a matter of fact, genes with similar functions are likely to be regulated via the same mechanisms and show similar expression patterns [15]. For instance, both the genes *TP53* and *MDM2* are verified to share the same function in regulating cell cycle and apoptosis, and thus we can observe their increased expression values in tumorigenic cells[22, 37]. Based on the above considerations, we utilize the graph learning to consider the gene-gene association by incorporating the prior gene ontology (GO) knowledge [6]. The GO is actually a semantic database, which provides a series of GO Terms to depict the attributes of genes. Generally, all the Go Terms can be divided into 3 different biological domains, (*i.e.,* Biological Process (BP), Molecular Function (MF), and Cellular Component (CC)) [6]. Here, each domain can be represented by a directed acyclic graph (DAG), with broad terms at the top and more specific terms as moving down (More details are presented in *Supplementary Materials*). In what follows, we will present a novel similarity measurement between pairs of genes based on the DAG of GO.

Considering two genes, $g_i$ and $g_j$, which can be represented by $n_i$ and $n_j$ different GO Terms. We use Eq.(10) to compute the pairwise genes similarity matrix $S^g$ whose elements are:

$$S_{ij}^g = \frac{1}{n_i n_j}\sum_{m=1}^{n_i}\sum_{k=1}^{n_j} \mathit{Sim}(\text{GO}_m, \text{GO}_k) \tag{10}$$

where,

$$\mathit{Sim}(\text{GO}_m, \text{GO}_k) = \begin{cases} b_{m,k} & \text{if GO}_m, \text{GO}_k \text{ in the same domain,} \\ 0 & \text{otherwise,} \end{cases} \tag{11}$$

and

$$b_{m,k} = \frac{\mathit{Intersect}(\text{GO}_m, \text{GO}_k)}{\mathit{max}\,(\mathit{Len}(\text{GO}_m), \mathit{Len}(\text{GO}_k))} \tag{12}$$

As shown in Eq.(10), the similarity between different genes can be viewed as the sum of their corresponding Go Terms. Specifically, if two Go Terms exist in the same domain (*i.e.,* BP, MF and CC), the similarity between them will be set according to Eq.(12). In Eq.(12), intersect($GO_m$,$GO_k$) counts the number of shared parent nodes between $GO_m$ and $GO_k$ in Gene Ontology, and $\mathit{Len}(GO_m)$ denotes the length of the path from root to $GO_m$.

Then, based on the pairwise gene similarity information, we propose to learn inter-dependent gene predictors $P = [p_1, p_2, \cdots, p_C]$ via GAT, where $C$ indicates the number of genes that need to be predicted. Specifically, for the purpose of predicting genes expression from $N$ spots represented as $Z \in R^{N\times C}$, we firstly construct a graph $G_g$ whose nodes correspond to genes and its adjacent matrix $A_g \in R^{C\times C}$ can be calculated via a threshold $\theta$. Here, $\theta$ is determined by averaging all the 10 largest similarity values for each gene according to $S^g$, and we assume $A_g^{ij} = 1$ if $S_{ij}^g > \theta$, otherwise $A_g^{ij} = 0$. Next, we use a single GAT layer to take the gene representation from pretrained Gene2Vec model [8] as inputs, and the output is $P \in R^{C\times v_s}$ with $v_s$ denoting the dimensionality of spot image representation, and thus the predicted genes expression for all the spots can be denoted as:

$$\hat{Z} = HP^T \tag{13}$$

Table 1. Average Pearson Correlation Coefficient (PCC) of predicted expression for top 50 highly expressed genes (HEG) and top 50 highly variable genes (HVG) compared to ground truth expressions. The '*' symbol indicates that our M2TGLGO is significantly better than the comparing methods.

| | DLPFC | | ccRCC | | BC | |
|---|---|---|---|---|---|---|
| | HVG | HEG | HVG | HEG | HVG | HEG |
| Hist2ST[42] | 0.109 ± 0.0691 * | 0.052 ± 0.0824 * | 0.097 ± 0.0540 * | 0.081 ± 0.0026 * | 0.093 ± 0.0821 * | 0.085 ± 0.1465 * |
| HisToGene[27] | 0.116 ± 0.0274 * | 0.063 ± 0.0607 * | 0.121 ± 0.0059 * | 0.062 ± 0.1843 * | 0.102 ± 0.0805 * | 0.091 ± 0.0184 * |
| IHItoGene[18] | 0.124 ± 0.0523 * | 0.095 ± 0.1319 * | 0.143 ± 0.1832 * | 0.158 ± 0.0175 * | 0.191 ± 0.0644 * | 0.118 ± 0.0514 * |
| ST-Net[14] | 0.210 ± 0.0127 * | 0.117 ± 0.0555 * | 0.105 ± 0.0557 * | 0.113 ± 0.0171 * | 0.219 ± 0.0582 * | 0.207 ± 0.1009 * |
| TRIPLEX[5] | 0.236 ± 0.0230 * | 0.154 ± 0.0898 * | 0.251 ± 0.0934 * | 0.229 ± 0.0592 * | 0.246 ± 0.0402 * | 0.205 ± 0.0428 * |
| TG-GATEs[17] | 0.241 ± 0.0209 * | 0.186 ± 0.0932 * | 0.263 ± 0.0601 * | 0.174 ± 0.0977 * | 0.254 ± 0.0327 * | 0.149 ± 0.0936 * |
| BLEEP[39] | 0.252 ± 0.0617 * | 0.173 ± 0.0920 * | 0.254 ± 0.0574 * | 0.209 ± 0.0312 * | 0.227 ± 0.0775 * | 0.223 ± 0.0344 * |
| **M2TGLGO** | **0.291 ± 0.0218** | **0.217** ± **0.0431** | **0.296** ± **0.0433** | **0.256** ± **0.0489** | **0.308** ± **0.0501** | **0.269** ± **0.0409** |

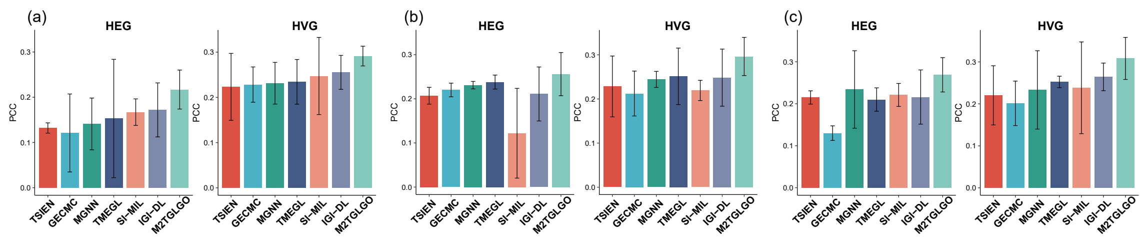


Figure 3. Comparison of M2TGLGO with other image representation methods on (a) DLPFC, (b) ccRCC and (c) BC datasets by the measurement of average PCC.

Where $H$ corresponds to the spots representation indicated in Eq.(6) and $\hat{Z} \in R^{N \times C}$. Finally, the mean-squared loss is applied to approximate the predicted gene expression values to the ground truth that can be formulated as follows:

$$\mathcal{L}_{\text{gene}} = \left\| Z - \hat{Z} \right\|_F \tag{14}$$

By the combination of Eq.(8), Eq.(9) and Eq.(14), the objective function of our M2TGLGO model can be represented as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{gene} + \mathcal{L}_s + \mathcal{L}_{recon} \tag{15}$$

## 4. Experimental Results

**Datasets:** In this study, we conduct our experiment on three public available ST datasets *i.e.,* DLPFC [23], BC [1], and ccRCC [24]. Specifically, the DLPFC dataset is generated from human brain samples, which is consisted of 12 tissue slices from three human samples with four adjacent slices for each. The BC cohort is collected for generating a comprehensive atlas of breast cancer that is consisted of 12 tissue slides. Moreover, the ccRCC dataset contains 12 samples with paired high-resolution pathology images and spatial resolved gene expression data. For each slide in these three datasets, we list its involved spots and gene number in *Supplementary Materials.*

**Experiment Setup and Evaluation Metrics:** For each dataset, we adopt the leave-one-out strategy(LOOCV) to test the performance of our method. Specifically, for each round of LOOCV, each slide is treated as a single test case, while the remaining samples are used for model training with 500 epoches. In the training process of our method, we adopt the AdamW optimizer with a learning rate of $1 \times 10^{-4}$ and the batch size is set as 1. For parameter settings, the MMGE module presented in section 3.3 is consisted of two GAT layers whose dimensions are 256 and 512, respectively. The dimension of gene representation from pre-trained Gene2Vec model is 200, and then will be transformed to the dimensionality of 512 after a GAT layer. In addition, we tune the parameter $\gamma$ in Eq.(5) from 0 to 1 with an interval of 0.2. We follow the study in [39] that evaluate the performance for gene expression prediction on the top 50 most highly variable genes (*i.e.,* HVG) and the top 50 most highly expressed genes(*i.e.,* HEG) by the measurements of Pearson Correlation Coefficient (PCC) and Mean Squared Error (MSE). Due to the page limitations, we only compare the PCC values of different methods in the main context, and the index of MSE are reported in the *Supplementary Materials* with consistent conclusions.

**Comparing Methods:** For the purpose of testing the effectiveness of our M2TGLGO, we compare it with the following *seven spatial resolved gene prediction methods:* Hist2ST[42], HisToGene[27], THItoGene[18], ST-Net[14], BLEEP[39], TRIPLEX[5], TG-GATEs[17],

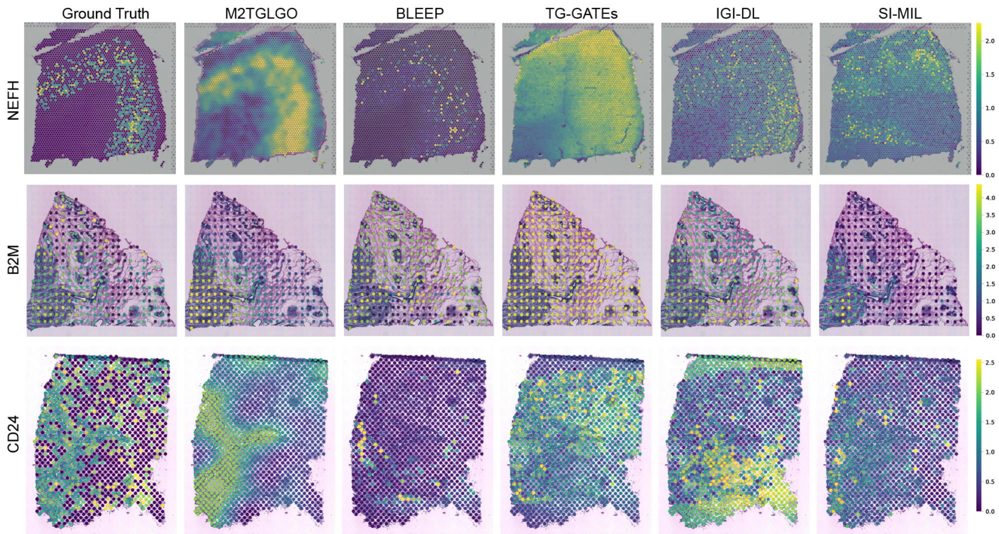


Figure 4. Comparison of the visualization results for spatial resolution gene expression.

as well as *six pathology image representation methods*:IGI-DL[11],SI-MIL[19], TMEGL[32], MGNN[10], GECMC[38], TSIEN[34]. More descriptions of these comparing methods are presented in *Supplementary Materials*.

**Comparison with ST based Gene Prediction Methods:** We first compare the gene prediction performance of M2TGLGO with the existing ST based gene prediction models on different datasets. As shown in Table 1 and Table **??** (provided in the *Supplementary Materials*), our method achieves significantly higher PCC as well as lower MSE than its competitors on all of the three datasets. The reason lies in the following two aspects: 1) We combine multi-modal pathological image features with the spatial information to generate better representations of spots. 2) We take the biological similarities among different genes into consideration, which is beneficial to the individual gene prediction tasks.

**Comparison with Pathology Image Representation Methods:** To further verify the superiority of the proposed multi-modal graph embedding method on spots, we also compare it with SOTA pathology image representation algorithms. The results shown in Fig. 3 and Fig. **??** (provided in *Supplementary Materials*) suggest that our spot representation method can best predict the gene profile from the ST data. More specifically, our M2TGLGO performs better than TMEGL[32], MGNN[10], GECMC[38] since we co-learn the spot representation from multi-modal features that can more comprehensively describe the pathology image. On the other hand, our method can better predict gene expression than the comparing IGI-DL [11], SI-MIL[19], GECMC[38] and TSIEN[34]. This is because our M2TGLGO designs the spatial-oriented ranking module that can preserve the neighborhood similarity information for spots representation learning.

**Comparison of the Visualization Results for Spatial Resolution Gene Expression:** To gain further insight into the predicted gene expression, we also visualize the spatial resolution gene expression on the histopathological images in Fig. 4. Specifically, for the marker gene *NEFH* on the DLPFC dataset (first row in Fig. 4), our method can predict its strong expression pattern on the cortical layer of the brain tissues, which is consistent with the ground truth and the existing knowledge[13]. Additional results on the ccRCC and BC datasets (second and third rows in Fig. 4) also indicate that our method can better infer the spatial distribution of different genes than its comparing methods, which again validates its effectiveness for gene expression prediction on ST data.

**Comparison of Different Methods for Gene-Gene Correlation Analysis:** For the gene expression prediction task, another way to evaluate its performance is to investigate if the predicted Gene-Gene Correlations (GGCs) can recover their original association[39]. Here, GGCs measure how the expression levels of one gene correlate with the expression levels of another across spatial locations that can reveal co-expression patterns, regulatory interactions, and functional relationships among different genes. We show the heatmap of Gene-Gene Correlations calculated from the predicted expressions of different methods in Fig. 5. As illustrated in Fig. 5, our proposed M2TGLGO can more accurately preserve the GGCs than its competitors, which clearly shows its potential in gaining a deeper un-

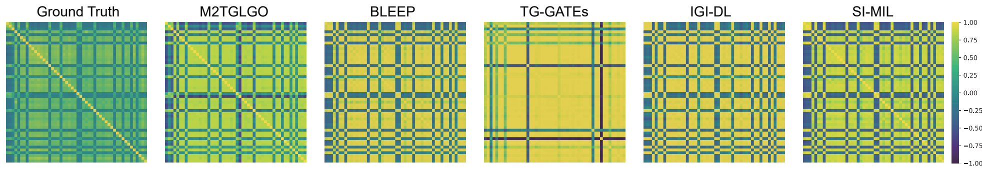


Figure 5. The heatmap of Gene-Gene Correlations calculated from the predicted expressions of each method on DLPFC dataset

derstanding of how gene interactions vary across different spatial locations.

Table 2. Ablation studies for different Modules in our M2TGLGO.

| **MMGE** | | **SONRM** | **GO Prior Knowledge** | **DLPFC** | | **ccRCC** | | **BC** | |
|---|---|---|---|---|---|---|---|---|---|
| Within-Modal | Inter-Modal | | | HVG | HEG | HVG | HEG | HVG | HEG |
| | ✓ | ✓ | ✓ | 0.090 | 0.073 | 0.215 | 0.183 | 0.162 | 0.115 |
| ✓ | | ✓ | ✓ | 0.236 | 0.182 | 0.229 | 0.204 | 0.260 | 0.213 |
| ✓ | ✓ | | ✓ | 0.252 | 0.187 | 0.258 | 0.222 | 0.261 | 0.232 |
| ✓ | ✓ | ✓ | | 0.163 | 0.102 | 0.216 | 0.160 | 0.117 | 0.223 |
| ✓ | ✓ | ✓ | ✓ | **0.291** | **0.217** | **0.296** | **0.256** | **0.308** | **0.269** |

**Ablation Study:** We conduct the ablation study to verify the effectiveness of MMGE and SONRM modules as well as the GO knowledge guided GNN for gene prediction. In the MMGE module, we further analyze the effects of within-modal and inter-modal embeddings, and the results are shown in Table 2. The proposed M2TGLGO (shown in the last row of Table 2) can better predict gene expression than all its competitors, indicating that each component contributes positively to the final prediction results.

**Effects of Parameter $\gamma$ in MMGE Module:** We also discuss the effects of parameter $\gamma$ that balances the contributions of within-modal and inter-modal embeddings. Specifically, we tune $\gamma$ from 0 to 1 with the interval 0.2 and the results are shown in Fig. 6. Most of inner intervals of the curves for both have larger PCC values (*i.e.,* better gene prediction performance) than the two vertices, which indicates the effectiveness of combining two types of embeddings for gene profile prediction.

**Effects of Combining Multi-modal Features:** To further verify the advantage of our method combining both handcrafted and deep features for gene expression prediction, we also compare it with its variations relying on partial modalities. As can be seen from Table **??** (provided in *Supplementary Materials*), our method (shown in the first row of Table **??**) outperforms its competitors, which demonstrates that each modality feature is valuable in predicting the genomic profiles on ST data.

**Effects of Quantile Values in the SONRM Module:** In our SONRM module, we set the quantile value as 4. In this section, we further discuss its effects from 3 to 6 with interval 1 and the results are shown in Table 3. We observe that the PCC values fluctuate in small ranges, indicating that our model is robust to the quantile values.

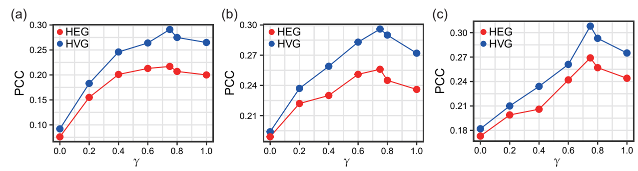


Figure 6. Performance of the model with varying values of hyper-parameter $\gamma$ on (a) DLPFC, (b) ccRCC, and (c) BC datasets by the measurement of PCC.

Table 3. Performance comparison with different Quantile values by the measurements of PCC.

| **Quantile** | **DLPFC** | | **ccRCC** | | **BC** | |
|---|---|---|---|---|---|---|
| | HVG | HEG | HVG | HEG | HVG | HEG |
| 3 | 0.285 | 0.210 | 0.298 | 0.231 | 0.300 | 0.257 |
| **4** | **0.291** | **0.217** | **0.296** | **0.256** | **0.308** | **0.269** |
| 5 | 0.289 | 0.204 | 0.281 | 0.240 | 0.295 | 0.258 |
| 6 | 0.292 | 0.211 | 0.288 | 0.241 | 0.302 | 0.260 |

## 5. Conclusion

In this paper, we propose a multi-modal topology-embedded graph Learning algorithm to help impute the gene expression of the queried spots from their corresponding pathological images. The main advantage of this study lies in the following two aspects: 1) Our proposed method can effective integrate multi-modal pathological image features by considering the topological structure among different spots. 2) We consider the correlation among different genes with prior gene ontology knowledge and develop a novel GNN model for predicting gene expressions in ST data. The experimental results on three public available ST datasets indicate that our proposed M2TGLGO can achieve significant higher prediction accuracy than all comparing methods, which highlights the potential of our method in bridging the gaps between tissues and sequencing in ST data.

## Acknowledgments

This work is supported by the National Natural Science Foundation of China (Nos.62136004, 62272226, 62402219).

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